

Land Surface Temperature Downscaling Using TsHARP and Machine Learning: A Comparative Study of Random Forest and XGBoost with SHAP-Based Explainability in Lucknow, India

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Abstract:

Land Surface Temperature (LST) is an important indicator of urban thermal conditions, but the spatial resolution of thermal satellite observations is generally coarser than that of multispectral optical imagery. This study investigates the downscaling of Landsat-8 LST from 90 m to 30 m over Lucknow, Uttar Pradesh, India, using a traditional thermal sharpening approach and machine learning models. TsHARP was implemented as a conventional statistical benchmark, while Random Forest (RF) and Extreme Gradient Boosting (XGBoost) were trained using six 30 m remote-sensing predictors: Green reflectance, SWIR1, SWIR2, NDVI, NDBI, and MNDWI. A spatial block-based train-test strategy was adopted for the machine learning models to reduce the potential bias associated with spatial autocorrelation. Model outputs were evaluated using mean absolute error (MAE), root mean square error (RMSE), coefficient of determination (R^2), coarse-scale consistency, spatial differences, and thermal-range characteristics. RF achieved an MAE of 0.742 °C, RMSE of 0.968 °C, and R^2 of 0.684, while XGBoost achieved an MAE of 0.741 °C, RMSE of 0.970 °C, and R^2 of 0.683, indicating nearly equivalent predictive performance. The two machine learning products also showed strong spatial similarity, with a mean absolute difference of 0.175 °C. However, both machine learning approaches produced narrower thermal ranges than the original 90 m LST, indicating a tendency toward thermal-range compression. SHAP-based analysis further showed that SWIR2, NDBI, and NDVI were among the dominant predictors, with higher NDVI generally contributing negatively and higher SWIR2 and NDBI generally contributing positively to model predictions. The results demonstrate that machine learning can provide spatially detailed LST estimates while also offering an interpretable framework for investigating predictor contributions. The study highlights both the potential and limitations of machine-learning-based LST downscaling in a heterogeneous urban environment.

Keywords:

Land Surface Temperature; LST Downscaling; Thermal Sharpening; TsHARP; Random Forest; XGBoost; Machine Learning; Remote Sensing; Landsat-8; SHAP; Explainable AI; Urban Heat Island

1. Introduction

Land Surface Temperature (LST) is an important indicator of the thermal characteristics of the Earth's surface and has wide applications in urban climate analysis, environmental monitoring, and assessment of surface thermal conditions [1], [2]. Satellite remote sensing provides an effective means of observing LST over large areas, but thermal observations generally have coarser spatial resolution than multispectral optical observations. This difference in spatial resolution limits the ability of satellite-derived LST to represent fine-scale thermal variability within heterogeneous urban environments [3].

LST downscaling aims to overcome this limitation by using relationships between coarse-resolution thermal observations and finer-resolution surface characteristics. Among traditional approaches, TsHARP uses the relationship between land surface temperature and vegetation-related information to generate higher-resolution thermal imagery [5],[6]. Although TsHARP has been widely applied, its performance can be affected by land-cover heterogeneity and the strength of the relationship between temperature and the selected sharpening variables [5].

Machine learning provides an alternative approach because it can simultaneously use multiple high-resolution predictors and model nonlinear relationships between surface characteristics and LST. Random Forest (RF) and Extreme Gradient Boosting (XGBoost) are two tree-based machine-learning methods that are well suited to heterogeneous remote-sensing data. Recent studies have demonstrated the potential of machine learning for producing high-resolution urban LST products from multispectral and thermal observations [9], [10]. However, predictive accuracy alone does not explain how individual environmental predictors influence model outputs.

Explainable artificial intelligence can provide additional insight into machine-learning models. SHapley Additive exPlanations (SHAP) assigns contribution values to input features and allows the magnitude and direction of their influence on individual predictions to be examined [11]. This provides an opportunity to investigate whether different machine-learning models learn similar relationships from remote-sensing predictors even when their predictive performance is comparable.

Therefore, this study presents a comparative assessment of TsHARP, Random Forest, and XGBoost for downscaling Landsat-8 LST from 90 m to 30 m over Lucknow, Uttar Pradesh, India. Six 30 m remote-sensing predictors—Green, SWIR1, SWIR2, NDVI, NDBI, and MNDWI—are used for the machine-learning models. The approaches are evaluated using spatial validation, coarse-scale consistency, spatial differences, and thermal-range characteristics. SHAP analysis is further used to interpret the contributions of the predictors to the RF and XGBoost predictions. The study aims to evaluate not only predictive performance but also the spatial and thermal behaviour and interpretability of the resulting downscaled LST products.

2. Materials and Methods

2.1 Study Area and Satellite Data

The study was conducted over Lucknow, Uttar Pradesh, India, an urban environment characterized by heterogeneous land-cover conditions including built-up surfaces, vegetation, bare land, and water bodies. This heterogeneity provides a suitable environment for evaluating spatial downscaling methods because different surface materials exhibit distinct thermal and spectral responses. Landsat-8 observations were used as the primary remote-sensing data source, comprising multispectral observations

from the Operational Land Imager (OLI) and thermal observations from the Thermal Infrared Sensor (TIRS). The Landsat Collection 2 Level-2 Surface Temperature product was used to obtain the reference land surface temperature, while the corresponding optical observations were used to derive higher-resolution spectral and spectral-index predictors. The Level-2 surface temperature product was used as the source thermal dataset and aggregated to a 90 m grid to construct the coarse-resolution LST reference, while the optical predictor variables were retained at 30 m spatial resolution for the subsequent 90 m-to-30 m downscaling experiments.

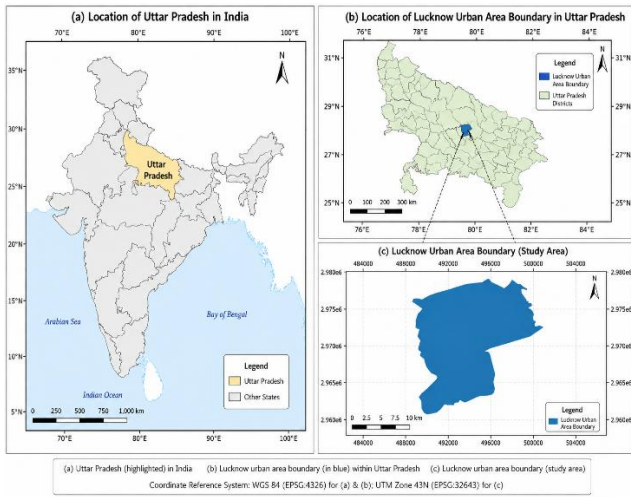


Figure 1. Location of the Lucknow study area in Uttar Pradesh, India. (a) Location of Uttar Pradesh in India; (b) Location of Lucknow in Uttar Pradesh; (c) Lucknow urban area boundary

2.2 Satellite Data

Landsat-8 satellite imagery was used as the primary remote sensing data source for the study. The dataset comprises multispectral observations from the Operational Land Imager (OLI) and thermal observations from the Thermal Infrared Sensor (TIRS). Landsat-8 Collection 2 Level-2 products were used, providing atmospherically corrected surface reflectance and surface temperature information. The thermal component

was used to obtain the Land Surface Temperature (LST), while the optical surface-reflectance bands were used to derive the spatial predictors required for LST downscaling. The predictor dataset consisted of Green reflectance, SWIR1 reflectance, SWIR2 reflectance, the Normalized Difference Vegetation Index (NDVI), the Normalized Difference Built-up Index (NDBI), and the Modified Normalized Difference Water Index (MNDWI). The OLI-derived predictors were retained at the 30 m spatial grid, while the Landsat-8 thermal product, although originating from the TIRS sensor with a native spatial resolution of approximately 100 m, was provided on the 30 m product grid. The resulting datasets were subsequently subjected to quality control, spatial alignment, and resolution preparation before being used in the TsHARP, Random Forest, and XGBoost downscaling experiments.

2.3 Retrieval of Land Surface Temperature

Land Surface Temperature (LST) for the Lucknow study area was obtained from the Landsat-8 Collection 2 Level-2 Surface Temperature product. The thermal surface temperature band ST_B10, derived from the Thermal Infrared Sensor (TIRS), was used as the primary LST data source. Rather than calculating temperature directly from at-sensor radiance using the Planck equation, the Level-2 surface temperature product was converted from its stored digital values using the temperature multiplicative scale factor and additive offset provided in the Landsat metadata. The metadata of the acquired scene specified a temperature multiplicative factor of 0.00341802 and an additive offset of 149.0 K.

For each pixel, the surface temperature in Kelvin was calculated as:

$$TK = DN_{ST} \times MST + AST$$

where represents the digital value of the ST_B10 pixel, is the temperature multiplicative scale factor, and is the temperature additive offset. For the Landsat-8 scene used in this study:

$$TK = DNST \times 0.00341802 + 149.0$$

The resulting temperature values were then converted from Kelvin to degrees Celsius using:

$$T^{\circ}C = TK - 273.15$$

Thus, the complete conversion applied in the study can be expressed as:

$$T^{\circ}C = (DNST \times 0.00341802 + 149.0) - 273.15$$

These equations directly correspond to the implementation in the LST retrieval notebook, where the temperature scale and offset are read from the Level-2 metadata before being applied to ST_B10.

Pixels with an ST_B10 value of zero, corresponding to the Collection 2 fill value, were treated as invalid and converted to NaN before further analysis. This prevented fill or invalid observations from being included in the temperature statistics and subsequent modelling.

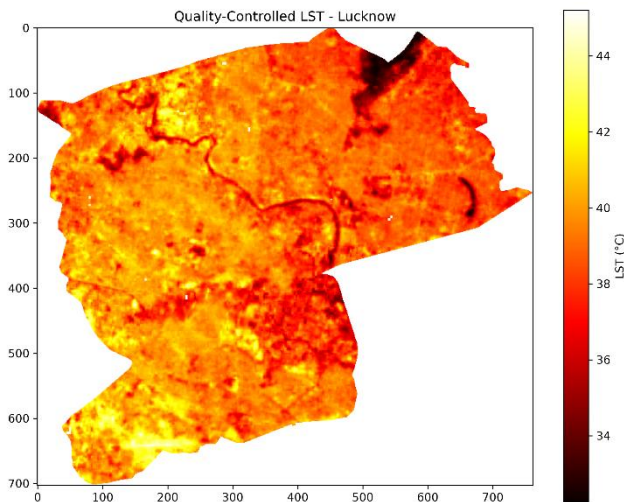


Figure 2.1 LST map of Lucknow

2.4 NDVI Computation

The Normalized Difference Vegetation Index (NDVI) was computed to quantify vegetation characteristics within the Lucknow study area. NDVI exploits the difference in spectral response between vegetation in the red and near-infrared (NIR) portions of the electromagnetic spectrum. Healthy vegetation strongly absorbs red radiation due to chlorophyll while exhibiting relatively high reflectance in the NIR region.

For the Landsat 8 Collection 2 Level-2 Surface Reflectance data used in this project, the red and NIR bands correspond to Band 4 (SR_B4) and Band 5 (SR_B5), respectively. NDVI was calculated on a pixel-by-pixel basis using:

$$NDVI = [(NIR - Red) / (NIR + Red)]$$

$$NDVI = [(SR_B5 - SR_B4) / (SR_B5 + SR_B4)]$$

where (SR_B5) represents the NIR surface reflectance and (SR_B4) represents the red surface reflectance.

The resulting NDVI raster provides a spatial representation of vegetation conditions across Lucknow. Higher NDVI values generally indicate greater vegetation abundance or vigor, whereas lower values correspond to sparse vegetation, built-up surfaces, bare land, or water. The index theoretically ranges from -1 to +1.

The computed NDVI layer was subsequently aligned with the other predictor variables used in the project. This ensured that the vegetation information was spatially consistent with the Landsat-derived land surface temperature and other spectral predictors used in the subsequent GeoAI modelling workflow.

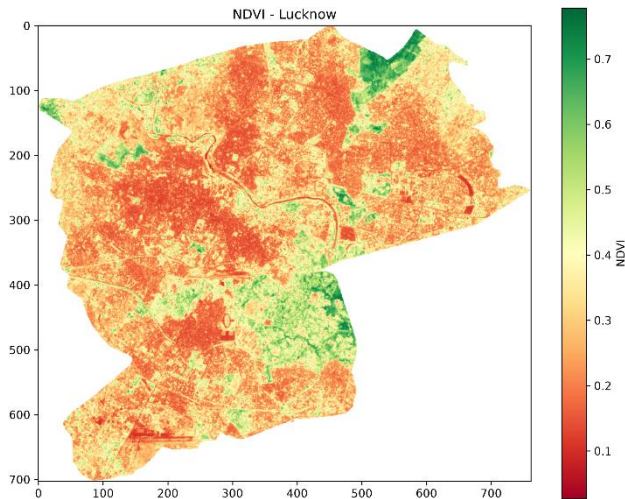


Figure 2.2 NDVI map of Lucknow

2.5 NDBI Computation

The Normalized Difference Built-up Index (NDBI) was computed to identify and quantify built-up characteristics within the Lucknow study area. NDBI utilizes the difference in spectral response between the Near-Infrared (NIR) and Short-Wave Infrared (SWIR) regions. Built-up surfaces generally exhibit relatively higher reflectance in the SWIR region compared with the NIR region, allowing the index to distinguish urban and built-up areas from vegetation and other land-cover types.

For the Landsat 8 Collection 2 Level-2 Surface Reflectance data used in this project, Band 5 (SR_B5) was used as the NIR band and Band 6 (SR_B6) as the SWIR1 band. NDBI was calculated for each pixel using:

$$NDBI = [(SWIR1 - NIR) / (SWIR1 + NIR)]$$

$$NDBI = [(SR_B6 - SR_B5) / (SR_B6 + SR_B5)]$$

where (SR_B6) represents the SWIR1 surface reflectance and (SR_B5) represents the NIR surface reflectance.

The resulting NDBI raster provides a spatial representation of the relative built-up

characteristics across Lucknow. Higher NDBI values generally indicate stronger built-up spectral characteristics, although NDBI may also respond to other spectrally similar non-vegetated surfaces. The index theoretically ranges from -1 to $+1$.

The computed NDBI layer was spatially aligned with the other predictor variables used in the project so that it could be incorporated consistently into the subsequent analysis and GeoAI modelling workflow.

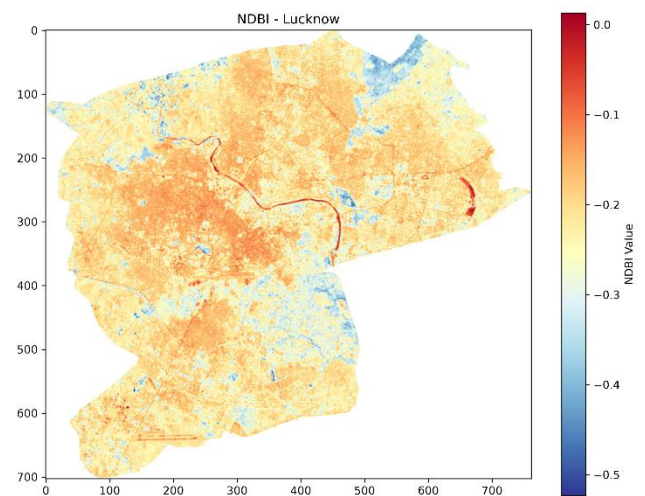


Figure 2.3 NDBI map of Lucknow

2.6 MNDWI Computation

The Modified Normalized Difference Water Index (MNDWI) was computed to enhance the detection of water bodies within the Lucknow study area. MNDWI is particularly useful in urban environments because it improves the separation of water surfaces from built-up areas by replacing the NIR band used in the conventional NDWI with the Short-Wave Infrared (SWIR1) band.

For the Landsat 8 Collection 2 Level-2 Surface Reflectance data used in this project, Band 3 (SR_B3) was used as the Green band and Band 6 (SR_B6) as the SWIR1 band. MNDWI was calculated on a pixel-by-pixel basis using:

$$\text{MNDWI} = [(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})]$$

$$\text{MNDWI} = [(\text{SR_B3} - \text{SR_B6}) / (\text{SR_B3} + \text{SR_B6})]$$

where (SR_B3) represents the Green surface reflectance and (SR_B6) represents the SWIR1 surface reflectance.

The resulting MNDWI raster enhances water features by producing relatively higher values for water surfaces, while built-up areas, vegetation, and bare surfaces generally produce lower values. Water surfaces generally exhibit higher MNDWI values, whereas built-up, vegetated, and bare surfaces tend to show lower values. The index theoretically ranges from -1 to $+1$.

The computed MNDWI layer was spatially aligned with the other remotely sensed predictor variables generated in the project. This enabled the water-related spectral information to be incorporated consistently into the subsequent spatial analysis and GeoAI modelling workflow.

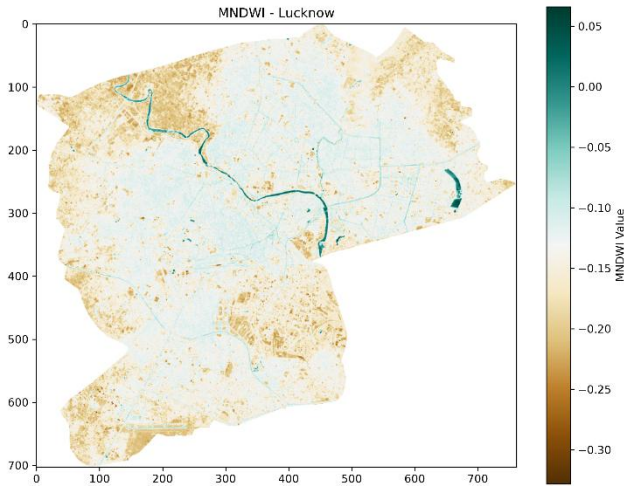


Figure 2.4 MNDWI map of Lucknow

2.7 TsHARP-Based Land Surface Temperature Downscaling

The TsHARP (Thermal Sharpening) approach was used as a conventional thermal downscaling

method to generate a 30 m LST product from the 90 m coarse-resolution reference LST. Unlike the machine-learning approaches, TsHARP used NDVI as the sole sharpening predictor. The method is based on the empirical relationship between land surface temperature and vegetation conditions across spatial scales [5], [6].

First, the 30 m NDVI layer was aggregated to the 90 m spatial resolution of the reference LST so that the temperature–vegetation relationship could be established at the coarse scale. The relationship between coarse-resolution LST and coarse-resolution NDVI was then modelled using regression:

$$\text{LST}_{(c)} = f(\text{NDVI}_{(c)}) + \varepsilon_{(c)}$$

where $\text{LST}_{(c)}$ represents the coarse-resolution reference LST, $\text{NDVI}_{(c)}$ represents NDVI aggregated to the coarse resolution, f represents the fitted temperature–NDVI relationship, and $\varepsilon_{(c)}$ represents the residual between the observed coarse-resolution LST and the regression-estimated LST.

The fitted temperature–NDVI relationship was subsequently applied to the original 30 m NDVI layer to obtain an initial fine-resolution LST estimate:

$$\text{LST}_{(f,\text{reg})} = f(\text{NDVI}_{(f)})$$

where $\text{NDVI}_{(f)}$ represents the 30 m NDVI and $\text{LST}_{(f,\text{reg})}$ represents the fine-resolution LST estimated from the regression relationship.

To preserve consistency with the coarse-resolution reference LST, the regression residual was then calculated at the coarse scale as:

$$\Delta \text{LST}_{(c)} = \text{LST}_{(c)} - \text{LST}_{(c,\text{reg})}$$

where $\text{LST}_{(c,\text{reg})}$ is the LST predicted by the regression model at the coarse resolution. The resulting residual field was transferred to the 30

m grid and added to the fine-resolution regression estimate:

$$LST(TsHARP) = LST(f,reg) + \Delta LST_{(c \rightarrow f)}$$

where $\Delta LST_{(c \rightarrow f)}$ represents the coarse-resolution residual field mapped to the 30 m target grid. This residual-correction step ensures that the downscaled product retains consistency with the original coarse-resolution thermal observations while incorporating the fine-scale spatial variability represented by NDVI. The use of a residual correction is a defining component of the TsHARP thermal-sharpening procedure [5], [6].

Invalid pixels were excluded using the prepared quality mask throughout the downscaling process. The resulting TsHARP product represents the estimated 30 m LST based on the temperature–NDVI relationship and the subsequent coarse-scale residual correction. The final product was then compared with the Random Forest and XGBoost downscaled LST products.

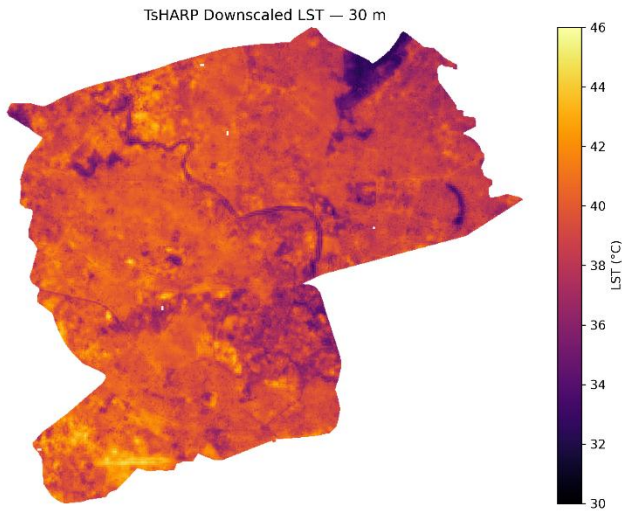


Figure 2.5 TsHARP downscaled LST map of Lucknow

2.8 Spatial Block-Based Train-Test Strategy

A spatial block-based train-test strategy was adopted for the Random Forest and XGBoost models to reduce the potential bias caused by spatial autocorrelation between neighbouring pixels. Instead of randomly distributing individual pixels between the training and testing datasets, the study area was divided into spatially distinct blocks.

The predictor values and corresponding coarse-resolution LST observations within the selected training blocks were used to train the machine-learning models, while spatially separate blocks were reserved for testing. The spatial separation reduced the likelihood of highly similar neighbouring observations occurring in both the training and testing datasets, providing a more realistic assessment of spatial generalization.

Only valid pixels remaining after application of the quality mask were included in the training and testing datasets. This approach reduced the possibility of neighbouring or highly similar pixels appearing in both datasets and provided a more robust evaluation of the spatial generalization of the RF and XGBoost models.

2.9 Random Forest-Based LST Downscaling

A **Random Forest (RF) regression model** was also used to downscale the coarse-resolution Land Surface Temperature (LST) to the target fine spatial resolution. Random Forest is an ensemble machine-learning algorithm that combines the predictions of multiple decision trees to model complex and non-linear relationships between input predictor variables and the target variable [7].

In this project, the coarse-resolution LST was used as the target variable, while the corresponding spectral and environmental variables were used as predictors. These

predictors included the previously generated **NDVI, NDBI, MNDWI, Green, SWIR1, and SWIR2** layers.

The basic relationship learned by the Random Forest model can be represented simply as:

Coarse LST = RF(NDVI, NDBI, MNDWI, Green, SWIR1, SWIR2)

The coarse-resolution predictor information was first matched spatially with the coarse LST observations. Pixels affected by invalid observations or quality issues were excluded using the prepared quality mask. The remaining valid pixels were used to train the Random Forest regression model.

After training, the model was applied to the corresponding fine-resolution predictor layers. The fine-resolution LST prediction can therefore be represented as:

Fine LST = RF(Fine NDVI, Fine NDBI, Fine MNDWI, Fine Green, Fine SWIR1, Fine SWIR2)

In this way, the Random Forest model learns the relationship between surface temperature and the spectral characteristics represented by the predictor variables at the coarse scale and transfers this relationship to the finer spatial resolution.

Random Forest was selected because it can capture non-linear relationships between LST and surface characteristics without requiring the relationship between the predictors and LST to be explicitly defined. It is also relatively robust to variations in predictor importance and complex interactions among input variables.

The resulting RF-downscaled LST raster provides a fine-resolution estimate of surface temperature. This output was subsequently evaluated against the available reference LST and compared with

the results obtained from the TSharp-based downscaling approach.

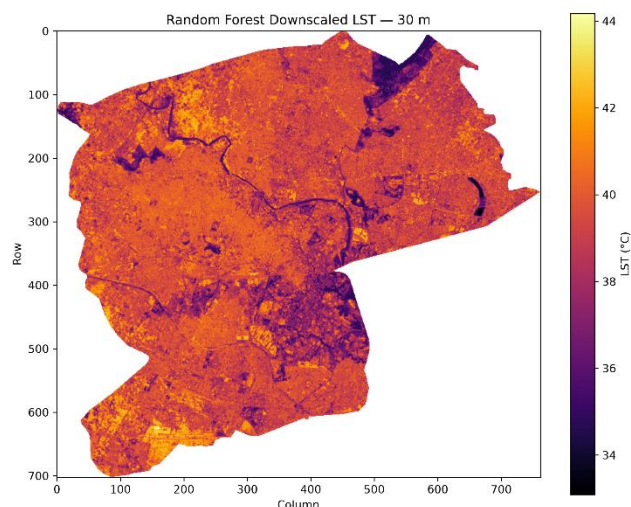


Figure 2.6 Random Forest Downscaled LST map of Lucknow

2.10 XGBoost-Based LST Downscaling

An **Extreme Gradient Boosting (XGBoost)** regression model was used as another machine-learning approach for downscaling Land Surface Temperature (LST). XGBoost is a gradient-boosting algorithm that builds a sequence of decision trees, where each successive tree attempts to improve the errors made by the previous trees. This allows the model to capture complex and non-linear relationships between surface temperature and the spectral characteristics of the study area [8].

In this project, the coarse-resolution LST was used as the target variable, while the corresponding predictor variables were used as input features. The predictor set consisted of **NDVI, NDBI, MNDWI, Green, SWIR1, and SWIR2**.

The relationship learned during model training can be represented simply as:

Coarse LST = XGBoost(NDVI, NDBI, MNDWI, Green, SWIR1, SWIR2)

Before model training, the coarse-resolution LST and predictor layers were spatially aligned. Pixels containing invalid observations were excluded using the prepared quality mask. The remaining valid pixels were used to construct the training dataset, with the predictor values forming the input features and the corresponding coarse-resolution LST values forming the target.

After training, the XGBoost model was applied to the fine-resolution predictor layers to estimate LST at the target spatial resolution:

Fine LST = XGBoost(Fine NDVI, Fine NDBI, Fine MNDWI, Fine Green, Fine SWIR1, Fine SWIR2)

The XGBoost approach is particularly useful for this application because the relationship between LST and surface characteristics is not necessarily linear. Vegetation, built-up surfaces, water bodies, and spectral properties can interact in complex ways, and the boosting framework can model such non-linear relationships.

The resulting XGBoost-downscaled LST raster represents the estimated fine-resolution spatial distribution of surface temperature.

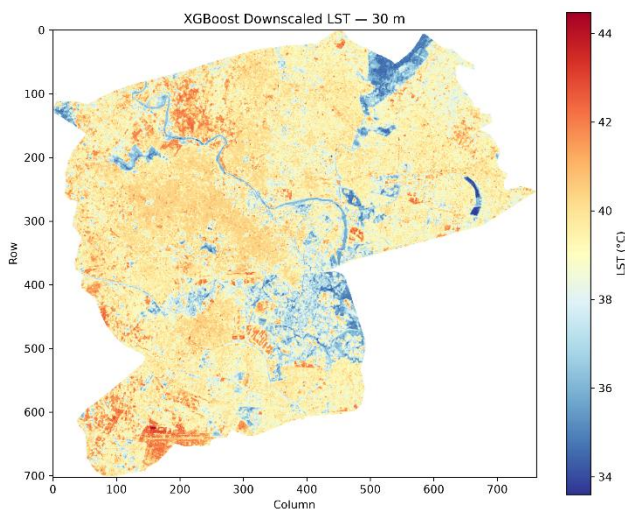


Figure 2.7 XGBoost Downscaled LST map of Lucknow

2.11 SHAP-Based Model Interpretation

SHapley Additive exPlanations (SHAP) was used to interpret the Random Forest and XGBoost models and to examine the contribution of individual predictors to the predicted LST. SHAP assigns a contribution value to each predictor for a given model prediction, allowing both the magnitude and direction of its influence to be analysed [11].

The six predictor variables—Green, SWIR1, SWIR2, NDVI, NDBI, and MNDWI—were analysed using SHAP values. The resulting feature contributions were used to identify the most influential predictors and to examine whether their effects on the predicted LST were positive or negative.

SHAP analysis was performed separately for the RF and XGBoost models to examine predictor contributions and compare their learned relationships.

2.12 Statistical Analysis and Coarse-Scale Consistency

Statistical analysis was performed to evaluate and compare the performance of the TsHARP, Random Forest, and XGBoost downscaled LST products. For the Random Forest and XGBoost models, predictive performance was evaluated using spatially withheld test observations and the metrics Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2). Bias was also calculated to assess the mean difference between predicted and reference LST.

In addition to spatial test performance, a coarse-scale consistency assessment was performed for all three downscaled products. Each 30 m LST product was aggregated back to the 90 m spatial

scale and compared with the original 90 m reference LST. MAE, RMSE, R^2 , and bias were calculated to evaluate how well the downscaled products preserved consistency with the original thermal observations after aggregation.

Spatial differences between the downscaled products and their thermal-range characteristics were also examined to assess differences in spatial behaviour and preservation of temperature variability. These complementary analyses were used to evaluate predictive performance, coarse-scale consistency, spatial similarity, and thermal-range preservation.

3. Results and Discussion

The developed downscaling framework was evaluated using the generated 30 m LST products and their consistency with the coarse-resolution reference LST. Random Forest (RF) and Extreme Gradient Boosting (XGBoost) models were evaluated using spatially withheld test data, while the downscaled products were further examined in terms of their spatial distribution, thermal range, and predictor contributions. The analysis also investigated the relationship between land-surface characteristics and LST to determine whether the machine-learning models captured physically meaningful spatial patterns.

3.1 Relationship between Land Indices and LST

The relationship between land-surface characteristics and LST was examined using the spectral and land-cover-related predictors used by the machine-learning models, with particular emphasis on NDVI and NDBI. The results indicate that vegetation and built-up characteristics contribute in opposite directions to the spatial variability of LST.

NDVI exhibited a predominantly negative relationship with predicted LST. Higher NDVI values were generally associated with negative SHAP values, indicating lower model-predicted LST relative to the model's baseline prediction. This behaviour is consistent with the expected thermal influence of vegetation, since vegetated surfaces can reduce land surface temperature through shading and evapotranspiration. In contrast, areas with sparse vegetation tend to have greater exposure of impervious or bare surfaces and therefore can exhibit higher surface temperatures. The SHAP analysis supports this interpretation, showing that higher NDVI values generally contribute negatively to the predicted LST.

NDBI showed the opposite tendency. Higher NDBI values, representing stronger built-up characteristics, generally contributed positively to predicted LST. This indicates that the model associated more strongly built-up areas with higher surface temperatures. Impervious materials such as concrete, asphalt, and rooftops typically have different thermal and energy-balance properties from vegetated surfaces and can promote greater surface heating. The positive contribution of NDBI indicates that the model associated stronger built-up spectral characteristics with higher predicted LST.

SWIR2 was also identified as one of the dominant predictors of LST, together with NDBI and NDVI. Its strong contribution indicates that shortwave-infrared information contains substantial spatial information relevant to the estimation of fine-resolution LST. SWIR reflectance is sensitive to surface properties including moisture, vegetation condition, and the characteristics of impervious and exposed surfaces. Consequently, the importance of SWIR2 indicates that the models also relied on spectral

information associated with surface properties relevant to spatial LST variation.

The SHAP dependence results for the principal predictors further demonstrate that the relationship between the predictors and LST is not necessarily a simple linear response. The contribution of each predictor varies across its observed range, indicating that the machine-learning models capture nonlinear interactions between surface characteristics. This is an important advantage of the tree-based models used in this study, because the relationship between remotely sensed spectral properties and LST is influenced simultaneously by vegetation, built-up surfaces, moisture conditions, and other spatial characteristics.

Overall, the predictor-response behaviour agrees with the expected physical interpretation of urban thermal environments: **vegetation tends to suppress surface temperature, whereas built-up and spectrally distinctive impervious surfaces tend to enhance it**. The consistency between the observed model relationships and the expected thermal behaviour provides qualitative support for the interpretability of the downscaling results.

However, these results should not be interpreted as proof of direct physical causation. SHAP values describe how the trained machine-learning model uses each predictor to generate its predictions; they do not independently establish causal relationships between a land-surface variable and LST. Furthermore, the SHAP analysis was performed on a sample of observations and should therefore be regarded as an interpretation of model behaviour rather than an exhaustive pixel-level statistical assessment.

The observed predictor relationships provide an important basis for interpreting the final downscaled products. The ability of the models to

associate high vegetation conditions with lower LST and stronger built-up characteristics with higher LST contributes to the preservation of spatial thermal contrasts at the 30 m scale.

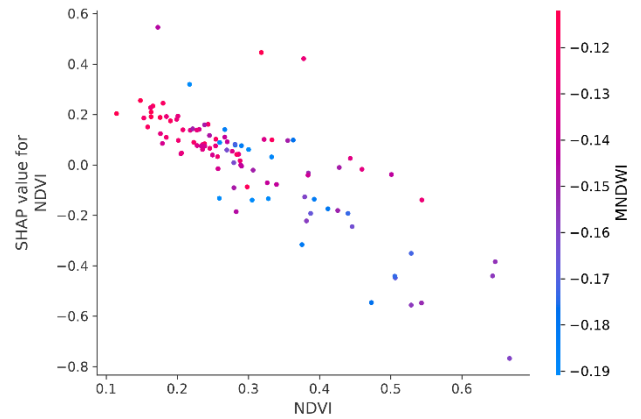


Figure 3.1. SHAP dependence plot showing the relationship between NDVI and predicted LST for the Random Forest model.

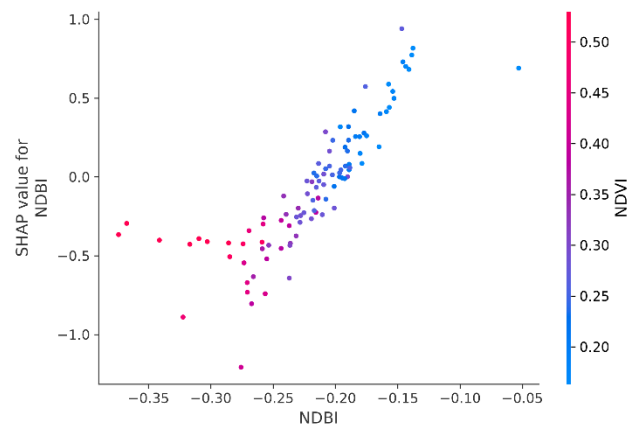


Figure 3.2. SHAP dependence plot showing the relationship between NDBI and predicted LST for the Random Forest model.

3.2 Performance of Machine-Learning Models

The Random Forest and XGBoost models demonstrated very similar predictive performance under the spatially withheld test dataset. Random Forest achieved a mean absolute error (MAE) of

0.742 °C, a root mean square error (RMSE) of **0.968 °C**, and an R^2 of **0.684**. XGBoost produced an MAE of **0.741 °C**, an RMSE of **0.970 °C**, and an R^2 of **0.683**.

The difference between the two models was therefore extremely small. XGBoost achieved a marginally lower MAE, indicating a slightly lower average absolute prediction error, whereas Random Forest achieved a marginally lower RMSE and slightly higher R^2 . These differences are too small to provide convincing evidence that one model substantially outperformed the other. Consequently, both models can be considered to have achieved comparable spatial generalization under the validation framework used in this study.

Model	MAE(°C)	RMSE (°C)	R^2
Random Forest	0.742	0.968	0.684
XGBoost	0.741	0.970	0.683

Figure 3.3 compares the prediction errors of the two machine-learning models, while **Figure 3.4** presents their corresponding R^2 values. The visual comparison is consistent with the numerical evaluation, showing that the performance of Random Forest and XGBoost is effectively tied.

An important observation is that the similar performance of the two models indicates that, under the present predictor set and validation framework, model choice between RF and XGBoost had only a small effect on predictive accuracy. Both models were trained using the same predictor set and spatial-validation strategy, allowing their performance to be compared under the same experimental conditions.

However, predictive accuracy alone does not fully determine the suitability of a downscaling model. A model may achieve a low average error while

failing to reproduce important spatial extremes or fine-scale thermal variability. Therefore, the subsequent analysis examines the spatial characteristics and thermal distributions of the generated 30 m LST products.

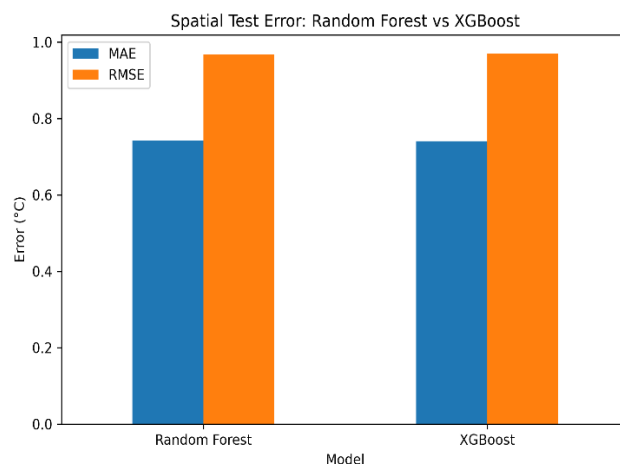


Figure 3.3. Comparison of MAE and RMSE for the Random Forest and XGBoost models.

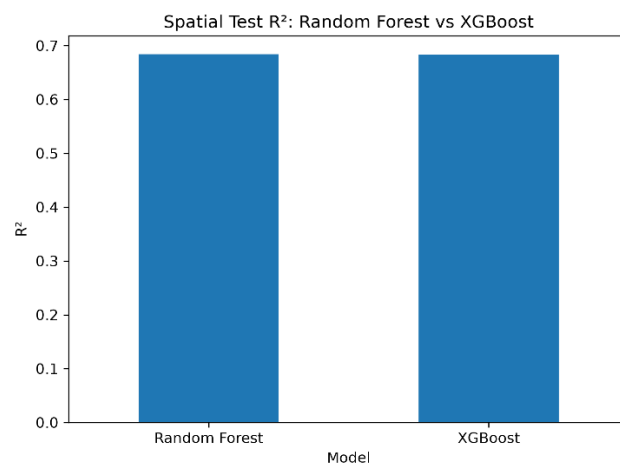


Figure 3.4. Comparison of coefficient of determination (R^2) for the Random Forest and XGBoost models.

3.3 Coarse-Scale Consistency

The coarse-scale consistency of the downscaled LST products was assessed by aggregating the 30 m products back to the original 90 m spatial scale and comparing them with the reference 90 m LST. The results are summarized in Table X.

TsHARP showed near-perfect coarse-scale consistency, with an MAE of approximately 0.000002 °C, RMSE of approximately 0.000003 °C, and R^2 of 1.000. This behaviour is expected because the TsHARP procedure includes a coarse-scale residual correction designed to preserve consistency with the original thermal observations.

Random Forest achieved a coarse-scale MAE of 0.637 °C, RMSE of 0.829 °C, and R^2 of 0.691, while XGBoost achieved an MAE of 0.643 °C, RMSE of 0.836 °C, and R^2 of 0.685. The results indicate that both machine-learning products retained substantial agreement with the original 90 m LST after aggregation. Random Forest showed slightly stronger coarse-scale agreement according to MAE, RMSE, and R^2 , although the difference between the two models was relatively small.

These results demonstrate that the machine-learning models were able to transfer the learned relationships to the 30 m scale while retaining a substantial degree of consistency with the original coarse-resolution thermal observations.

Model	MAE (°C)	RMSE (°C)	R^2	Bias (°C)
TsHARP	0.000002	0.000003	1.00	0.00
Random Forest	0.637	0.829	0.691	0.046
XGBoost	0.643	0.836	0.685	0.036

3.4 Spatial Characteristics of the Downscaled LST Products

The spatial characteristics of the generated 30 m LST products were examined to assess whether the downscaling approaches reproduced

meaningful fine-scale thermal patterns across the study area. Figure 3.5–3.7 present the 30 m LST products generated using TsHARP, Random Forest, and XGBoost, respectively. The three products reproduce the broad spatial structure of the thermal environment, while differences are observed in the magnitude and spatial variability of the estimated temperatures.

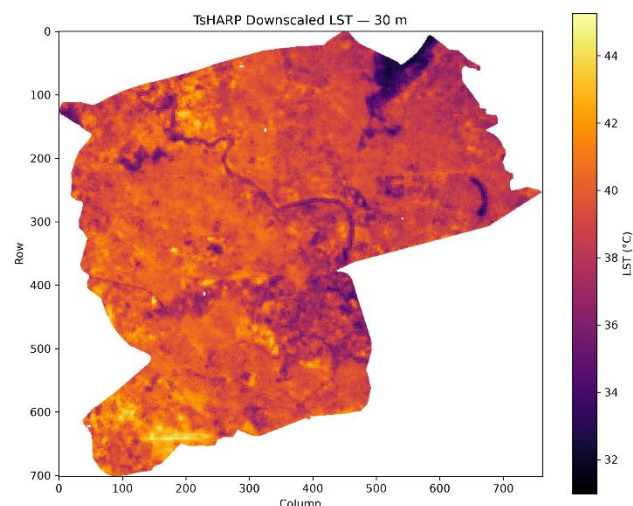


Figure 3.5. TsHARP-derived 30 m LST product for Lucknow.

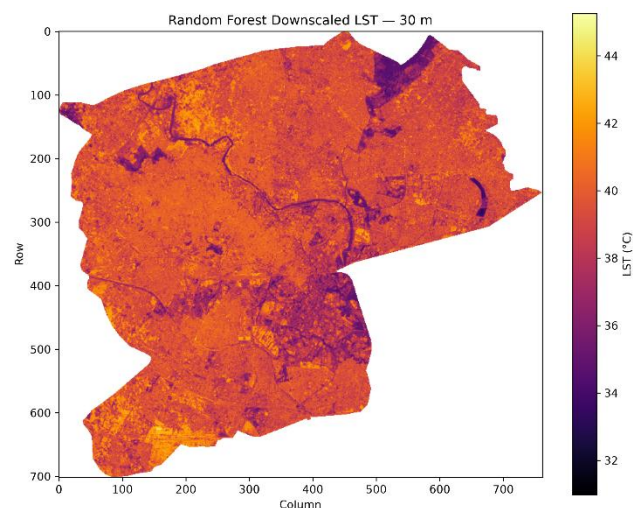


Figure 3.6. Random Forest-derived 30 m LST product for Lucknow.

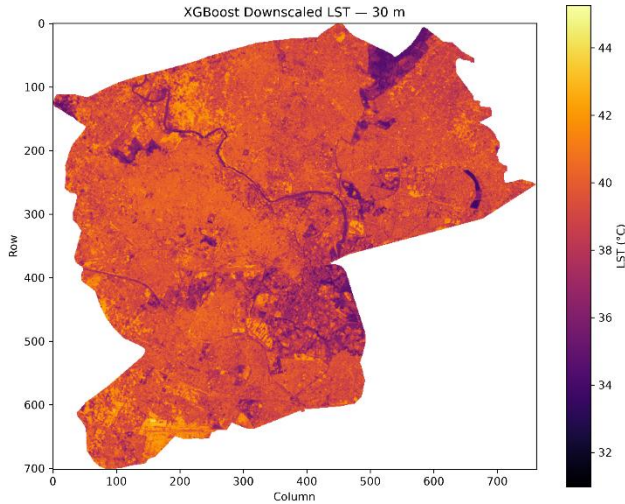


Figure 3.7. XGBoost-derived 30 m LST product for Lucknow.

3.5 Thermal Distribution and Extreme-Value Behaviour

The thermal distributions of the three downscaled LST products were compared to evaluate their ability to preserve temperature variability and extreme values. Figure 3.8 shows the distributions of the TsHARP, Random Forest, and XGBoost products.

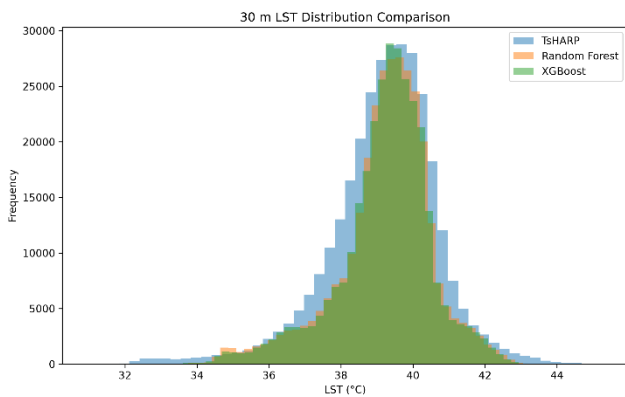


Figure 3.8. Comparison of LST distributions for the TsHARP, Random Forest, and XGBoost downscaled products.

Although the three approaches produce comparable mean thermal levels, their treatment of thermal variability differs considerably. TsHARP retains the broadest temperature range,

whereas Random Forest and XGBoost produce narrower distributions. This indicates that the machine-learning models capture the dominant spatial thermal pattern but tend to moderate the most extreme temperatures. Therefore, evaluation based only on mean temperature or average prediction error would not fully describe the differences between the downscaled products.

TsHARP produced the widest thermal range, from **30.98 to 45.25 °C**, with a standard deviation of **1.54 °C**. In comparison, Random Forest ranged from **33.09 to 44.17 °C** with a standard deviation of **1.35 °C**, while XGBoost ranged from **33.60 to 44.47 °C** with a standard deviation of **1.33 °C**. Despite these differences, the mean LST values were very similar: **39.18 °C** for TsHARP, **39.22 °C** for Random Forest, and **39.21 °C** for XGBoost.

The results indicate that the machine-learning models reproduce the overall thermal level but produce narrower temperature distributions than TsHARP, suggesting some regression toward the mean and reduced representation of extreme temperatures. However, the wider range of TsHARP should not be interpreted as evidence of higher accuracy because the three approaches were not evaluated using identical independent validation procedures.

3.6 Spatial Difference Analysis

The spatial differences between the downscaled LST products were analysed to quantify how closely the different approaches reproduced similar fine-resolution thermal patterns. Figure 3.9–3.11 show the spatial difference maps for Random Forest versus TsHARP, XGBoost versus TsHARP, and XGBoost versus Random Forest, respectively.

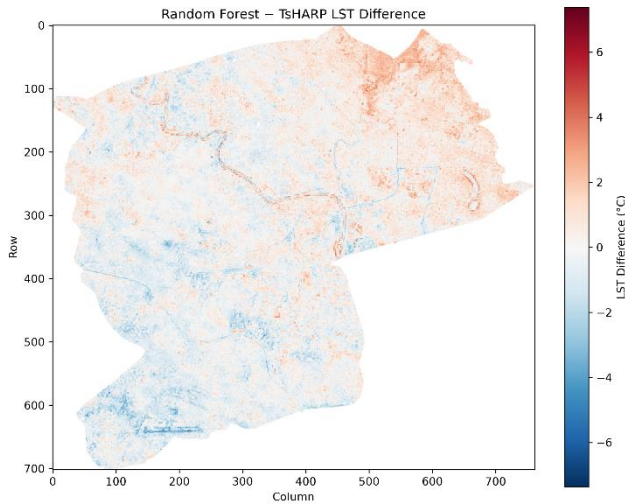


Figure 3.9. Spatial difference between the Random Forest and TsHARP 30 m LST products.

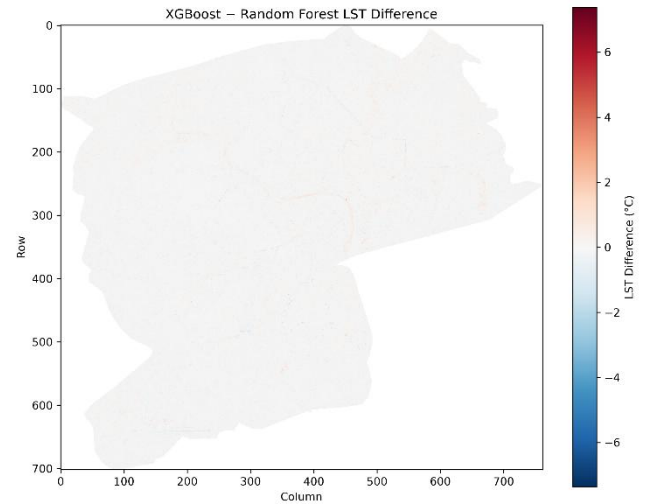


Figure 3.11. Spatial difference between the XGBoost and Random Forest 30 m LST products.

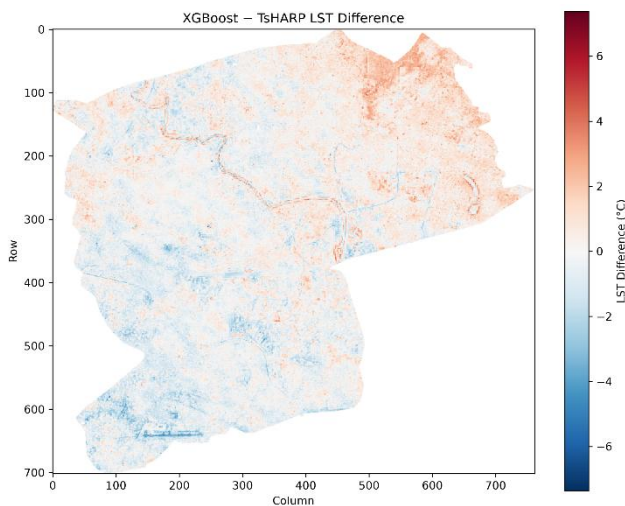


Figure 3.10. Spatial difference between the XGBoost and TsHARP 30 m LST products.

Overall, the spatial-difference analysis demonstrates that the two machine-learning models are substantially more consistent with each other than either is with TsHARP. This indicates that the two machine-learning approaches produced highly similar spatial estimates when trained using the same predictor set and validation framework.

The XGBoost and Random Forest products showed the strongest agreement, with a mean absolute spatial difference of approximately 0.175 °C. In comparison, the mean absolute differences between Random Forest and TsHARP and between XGBoost and TsHARP were approximately 0.74 °C. These results indicate that the two machine-learning approaches generated substantially more similar spatial thermal patterns than either approach did relative to TsHARP.

3.7 Overall Discussion

The results demonstrate that the machine-learning-based downscaling framework can generate fine-resolution LST products while preserving meaningful spatial relationships between land-surface characteristics and temperature. The SHAP analysis identified **NDVI**, **NDBI**, and **SWIR2** as the dominant predictors, with higher NDVI generally contributing to lower predicted LST and higher NDBI and SWIR2 contributing positively to predicted LST.

Random Forest and XGBoost achieved almost identical predictive performance, with MAEs of

0.742 °C and **0.741 °C**, respectively, and R^2 values of **0.684** and **0.683**. Therefore, the present results do not provide sufficient evidence to identify either algorithm as clearly superior.

Both machine-learning models also reproduced the overall mean thermal level of the study area closely, but produced narrower thermal distributions than TsHARP. This indicates a tendency to smooth extreme temperatures, which is an important limitation when the objective is to represent localized urban thermal hotspots. At the same time, the wider range produced by TsHARP cannot be interpreted as proof of greater accuracy because the approaches were not evaluated using identical independent validation procedures.

The close agreement between Random Forest and XGBoost, combined with their similar spatial products, suggests that the selected remote-sensing predictors contain substantial information for fine-resolution LST estimation. However, the remaining differences in thermal extremes indicate that further improvement may require predictors or modelling strategies capable of representing localized nonlinear thermal behaviour more effectively.

Overall, the results support the feasibility of machine-learning-based LST downscaling for producing spatially detailed thermal information, while also highlighting the need to carefully evaluate **extreme-value preservation, spatial generalization, and model interpretability** rather than relying solely on average prediction-error metrics.

4. Conclusion and Future Work

This study developed and evaluated a machine-learning-based framework for downscaling land surface temperature (LST) from a coarse-resolution reference to 30 m spatial resolution over Lucknow using remotely sensed land-

surface predictors. Random Forest and XGBoost were implemented and compared with the TsHARP approach to evaluate their predictive performance, spatial characteristics, thermal distributions, and ability to preserve fine-scale temperature variability.

The results demonstrated that Random Forest and XGBoost achieved comparable predictive performance under spatial validation. Random Forest obtained an MAE of **0.742 °C**, RMSE of **0.968 °C**, and R^2 of **0.684**, while XGBoost achieved an MAE of **0.741 °C**, RMSE of **0.970 °C**, and R^2 of **0.683**. The differences between the models were therefore negligible, indicating that neither model can be considered clearly superior based on predictive accuracy alone.

The interpretation of model behaviour showed that **NDVI, NDBI, and SWIR2** were the dominant predictors of LST. Higher NDVI generally contributed negatively to predicted LST, whereas higher NDBI and SWIR2 contributed positively. These relationships are consistent with the expected influence of vegetation and built-up surface characteristics on urban thermal conditions and indicate that the models learned meaningful relationships between spectral land-surface properties and LST.

The comparison of the generated 30 m products further revealed that the machine-learning models reproduced the overall thermal level of the study area closely but produced narrower temperature distributions than TsHARP. TsHARP retained the widest thermal range, whereas Random Forest and XGBoost showed greater concentration around the mean. This indicates that the machine-learning approaches have a tendency to smooth extreme temperatures. Although this behaviour provides relatively stable predictions, it may limit the representation of highly localized thermal hotspots.

The spatial difference analysis also showed strong agreement between the two machine-learning products, with a mean absolute difference of only **0.175 °C**, compared with approximately **0.74 °C** between either machine-learning product and TsHARP. This suggests that the selected predictors strongly constrain the resulting thermal patterns, allowing two different ensemble algorithms to produce broadly similar fine-resolution LST fields.

Despite these encouraging results, several limitations remain. The machine-learning models showed reduced representation of thermal extremes, and the current evaluation does not establish that the wider TsHARP temperature range is more accurate because the approaches were not evaluated using identical independent validation procedures. In addition, the SHAP analysis represents model behaviour rather than direct physical causation.

Future work should therefore focus on improving the representation of thermal extremes through additional spectral, environmental, and spatial predictors, as well as evaluating more advanced spatial machine-learning and GeoAI approaches. Testing the framework across multiple dates and seasons would also help determine its temporal generalizability. Most importantly, future evaluation should incorporate independent higher-resolution thermal observations where available to provide a stronger assessment of the spatial accuracy of the downscaled products.

Overall, the study demonstrates that machine-learning-based downscaling can provide a practical approach for generating fine-resolution LST information from coarser thermal observations. While Random Forest and XGBoost produced comparable and spatially consistent results, the observed compression of thermal extremes highlights an important area for further

methodological improvement. The findings provide a basis for developing more robust GeoAI-based approaches for urban thermal mapping and heat-environment analysis at fine spatial scales.

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